Waves at boundaries

antocciphysics | TEACHER KEY | CPA

Learning target: Describe reflection and transmission at a boundary.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 1; 3 course days later.*
Need wave speed; target distance=speed×3/2. The pulse travels to the target and back.

02. Explain a worked example (worked)
Fixed: inverted reflected pulse; free: same orientation, direction reversed.

03. Look and explain (model)
Downward/inverted pulse travels back along rope.

04. Build the solution (practice)
6 J.

05. Use the idea (reason)
No; source fixes frequency. Speed and wavelength may change; amplitude depends energy transfer and medium properties.

06. Choose a method (mixed) *Review: Unit 5, Day 4; 12 course days later.*
Q=mL=2400 J.

07. Exit check (exit)
Upright.

08. STRETCH: Apply it another way (stretch)
Displacements add during overlap; pulses subsequently pass through each other in the linear idealization.

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Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 1; 3 course days later.*
Second/first range=5/3≈1.667. Delay does not establish mass.

02. Explain a worked example (worked)
Fixed: inverted reflected pulse; free: same orientation, direction reversed.

03. Represent and interpret (model)
Downward/inverted pulse travels back along rope.

04. Solve and justify (practice)
Transmitted .50 J; reflected 25%, absorbed 12.5%, transmitted 62.5%.

05. Analyze the reasoning (reason)
No; source fixes frequency. Speed and wavelength may change; amplitude depends energy transfer and medium properties.

06. Choose a method (mixed) *Review: Unit 5, Day 4; 12 course days later.*
L=|Q|/m≈300000 J/kg; sample Q=-120000 J.

07. Exit check (exit)
2 m and 4 m.

08. STRETCH: Apply it another way (stretch)
Total zero at that point; energy may be in motion/elsewhere and interference redistributes it.